25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Systems of Linear Equations & Row Reduction Practice Exercises

1 MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21
B 14
C 15
D 225

2 MCQ

A 3×3 symmetric matrix Q_k has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26
B 32
C 29
D 41

3 MCQ

A linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

A $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$
B $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
C $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$
D $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

4 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

A (7, 6)
B (6, 7)
C (8, 5)
D (5, 8)

5 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

A 1
B 0
C -7
D 25

6 MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

A 1

B 2

C 3

D 4

7 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

A 13

B 17

C 7

D 169

8 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

B The angle the vector makes with the positive x-axis.

C The distance of the point (3,4) from the origin.

D The sum of the components of the vector.

9 MCQ

Consider the following system of linear equations represented by an augmented matrix: $\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \end{array}\right]$. What is the value of x in the first equation?

A 4

B 1

C 2

D 3

10 MCQ

Consider the upper-triangular matrix $W = \begin{pmatrix} 4 & 19 \\ 0 & 8 \end{pmatrix}$. Determine the eigenvalue product $\det(W) = \lambda_1 \lambda_2$.

A 36

B 35

C 32

D 13

11 MCQ

Solve for z: $x+y+z=-1$, $y+z=-4$, $z=1$ (already row-echelon).

A 1

B 3

C -5

D 2

12 MCQ

Solve for x: $x+y+z=6$, $y+z=-3$, $z=2$ (already row-echelon).

A -5

B 2

C 11

D 9

13 MCQ

Solve for x: $x+y+z=-4$, $y+z=4$, $z=-7$ (already row-echelon).

A 11

B -7

C -6

D -8

14 MCQ

Solve for z: $x+y+z=4$, $y+z=-3$, $z=-7$ (already row-echelon).

A 7

B 4

C -5

D -7

15 MCQ

Solve for y: $x+y+z=8$, $y+z=3$, $z=-5$ (already row-echelon).

A 5

B -5

C 10

D 8

16 MCQ

Solve for z: $x+y+z=-1$, $y+z=4$, $z=7$ (already row-echelon).

A 7

B 9

C -5

D -3

17 MCQ

Solve for y: $x+y+z=8$, $y+z=-7$, $z=-3$ (already row-echelon).

A -4

B 15

C -3

D -2

18 MCQ

Solve for z: $x+y+z=3$, $y+z=-8$, $z=4$ (already row-echelon).

A 11

B -12

C 6

D 4

19 MCQ

Solve for z: $x+y+z=-8$, $y+z=8$, $z=-4$ (already row-echelon).

A -16

B 12

C -4

D -2

20 MCQ

Solve for x: $x+y+z=3$, $y+z=1$, $z=5$ (already row-echelon).

A 2

B 5

C 4

D -4

21 MCQ

Find the root(s) satisfying for y: $x+y+z=3$, $y+z=2$, $z=-5$ (already row-echelon).

A 1

B 7

C -5

D 9

22 MCQ

Find the root(s) satisfying for z: $x+y+z=-2$, $y+z=-3$, $z=1$ (already row-echelon).

A 3

B 1

C 2

D -4

23 MCQ

Find the root(s) satisfying for z: $x + y + z = 2$, $y + z = -1$, $z = 3$ (already row-echelon).

A -4

B 3

C 5

D 4

24 MCQ

Solve for y: $x + y + z = -4$, $y + z = 4$, $z = 5$ (already row-echelon).

A -1

B -8

C 5

D 1

25 MCQ

Find the root(s) satisfying for x: $x + y + z = 4$, $y + z = 2$, $z = 6$ (already row-echelon).

A 4

B 2

C -4

D 6